

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

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COURSE NAME: Cloud Computing and Big Data Analytics

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Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Definition:

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, software, and analytics over the Internet on a **pay-as-you-go** basis. It enables users to access computing resources without owning or maintaining physical infrastructure.

Four Characteristics

1. On-Demand Self-Service

- Users can provision computing resources (VMs, storage, databases) whenever required without human intervention from the provider.
- **Example:** Creating an AWS EC2 virtual machine in a few minutes.

2. Broad Network Access

- Cloud services are accessible over the Internet using laptops, smartphones, tablets, or desktops from anywhere.
- **Example:** Accessing Google Drive or Microsoft OneDrive from any device.

3. Rapid Elasticity (Scalability)

- Resources can be increased or decreased automatically based on workload demand.

- **Example:** An e-commerce website automatically adds servers during a festival sale.

4. Measured Service (Pay-as-You-Go)

- Resource usage is monitored, and users pay only for what they consume.
- **Example:** Paying only for the storage used in Amazon S3.

Cloud Computing vs Traditional On-Premise Computing

Cloud Computing	Traditional On-Premise Computing
Resources available on demand	Hardware must be purchased and installed
Accessible from anywhere via the Internet	Usually limited to the local network
Easily scalable within minutes	Scaling requires buying and configuring new hardware
Pay only for actual usage	High upfront investment and maintenance costs

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Introduction

Cloud computing evolved through advances in computer hardware, networking, virtualization, distributed systems, and Internet technologies. These developments made modern cloud platforms such as AWS, Microsoft Azure, and Google Cloud possible.

Evolution of Cloud Computing

1. Hardware Evolution

- Early computing relied on expensive mainframes and dedicated physical servers.
- Improvements in processors, storage, and networking enabled powerful and affordable servers.
- **Example:** Multi-core processors and high-capacity SSDs increased computing performance.

2. Virtualization Technology

- Hypervisors allowed multiple virtual machines (VMs) to run on a single physical server.
- Improved hardware utilization, reduced costs, and simplified resource management.
- **Example:** VMware and Hyper-V.

3. Distributed Computing

- Multiple computers were connected to work together as a single system.
- Technologies such as clusters and grid computing improved scalability, reliability, and fault tolerance.
- **Example:** Hadoop clusters processing large datasets.

4. Internet and Software Evolution

- High-speed Internet, web services, APIs, and Service-Oriented Architecture (SOA) enabled applications to be accessed over the web.

- Users could access software and data from anywhere without local installation.
- **Example:** Gmail, Google Docs, and Microsoft 365.

How These Changes Enabled Modern Cloud Platforms

Evolution Stage	Contribution to Modern Cloud Platforms
Hardware Evolution	Powerful, reliable, and cost-effective servers
Virtualization	Efficient resource sharing and rapid VM creation
Distributed Computing	High availability, scalability, and fault tolerance
Internet Software Evolution	Web-based services, SaaS, APIs, and global accessibility

Examples

- **AWS** provides on-demand virtual servers (EC2) using virtualization.
- **Microsoft Azure** offers scalable cloud applications and storage.
- **Google Cloud Platform (GCP)** delivers global computing and AI services over the Internet.

Conclusion

The evolution from advanced hardware → virtualization → distributed computing → Internet software transformed traditional computing into modern cloud computing. These technologies enable today's cloud platforms to provide scalable, reliable, cost-effective, and globally accessible services

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Introduction

Server virtualization is the process of dividing one physical server into multiple Virtual Machines (VMs) using a hypervisor. Each VM operates independently with its own operating system and applications. It is the key technology that enables cloud computing.

Role of Server Virtualization in Cloud Computing

1. Efficient Resource Utilization

- Multiple VMs share the same physical server, reducing hardware wastage.
- **Example:** One server hosts Windows, Linux, and database servers simultaneously.

2. Resource Isolation

- Each VM runs independently, so failures in one VM do not affect others.
- Improves security and reliability.

3. Rapid Provisioning

- New virtual machines can be created, started, or deleted within minutes.
- Supports on-demand cloud services.

4. Scalability and Flexibility

- Resources such as CPU, memory, and storage can be allocated dynamically based on workload.

5. Cost Reduction

- Reduces the number of physical servers, lowering hardware, power, cooling, and maintenance costs.

Virtualization vs Cloud Computing

Virtualization (Enabling Technology)	Cloud Computing (Service Model)
A technology that creates multiple virtual machines on a physical server	A model for delivering computing services over the Internet
Uses hypervisors such as VMware, Hyper-V, or KVM	Provides IaaS, PaaS, and SaaS services
Focuses on efficient hardware utilization	Focuses on providing on-demand services to users
Can exist without cloud computing	Often uses virtualization to deliver cloud services
Example: Creating multiple VMs on one server	Example: AWS EC2, Microsoft Azure Virtual Machines

Examples

- **Virtualization:** VMware ESXi creates multiple virtual machines on a single server.
- **Cloud Computing:** AWS EC2 provides virtual servers on demand through the Internet using virtualization.

Conclusion

Server virtualization is the **foundation of cloud computing** because it enables efficient resource sharing, isolation, scalability, and rapid provisioning.

4. Discuss the role of data centers in delivering cloud services.

Introduction

A **data center** is a secure facility that houses physical servers, storage systems, networking devices, and power and cooling infrastructure. Data centers are the backbone of cloud computing because they host and manage cloud resources.

Role of Data Centers in Cloud Services

1. Resource Hosting

- Store and run cloud resources such as virtual machines, databases, storage, and applications.
- **Example:** AWS data centers host EC2 instances and Amazon S3 storage.

2. High Availability

- Redundant servers, backup power, and network connections ensure continuous service with minimal downtime.
- **Example:** If one server fails, another server automatically takes over.

3. Scalability

- Data centers allow cloud providers to quickly add servers and storage to meet increasing user demand.
- **Example:** During online shopping festivals, additional resources are allocated automatically.

4. Data Storage and Backup

- Store user data securely and maintain backups to prevent data loss.
- Supports disaster recovery and business continuity.

5. Security

- Protect data using physical security (CCTV, biometric access), firewalls, encryption, and continuous monitoring.
- Ensures confidentiality, integrity, and availability of data.

6. Virtualization Support

- Data centers use virtualization to create multiple virtual machines on physical servers, improving resource utilization and reducing costs.

Examples

- **Amazon Web Services (AWS):** Global data centers provide computing, storage, and networking services.
- **Microsoft Azure:** Data centers deliver virtual machines, databases, and AI services worldwide.
- **Google Cloud Platform (GCP):** Uses globally distributed data centers for reliable cloud services.

Conclusion

Data centers are the foundation of cloud computing. They provide the infrastructure required for hosting applications, storing data, ensuring security, supporting virtualization, enabling scalability, and delivering reliable cloud services to users around the world.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Introduction

Cloud service models define how computing resources are delivered to users. The main models are Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS), and Anything as a Service (XaaS).

Comparison of Cloud Service Models

Service Model	Description	What the Provider Offers	Real-World Example
IaaS (Infrastructure as a Service)	Provides virtual servers, storage, and networking over the Internet. Users manage the OS and applications.	Virtual machines, storage, networking	Amazon EC2 (AWS)
PaaS (Platform as a Service)	Provides a platform for developing, testing, and deploying applications without managing hardware.	Runtime environment, databases, development tools	Google App Engine
SaaS (Software as a Service)	Delivers ready-to-use software over the Internet through a web browser.	Complete software applications	Google Docs

XaaS (Anything as a Service)	A broad model that delivers various IT services through the cloud, such as storage, security, databases, AI, and networking.	Any cloud-based service	Amazon S3 (Storage as a Service)
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Key Differences

- IaaS provides infrastructure such as virtual machines and storage.
- PaaS provides a development platform for building applications.
- SaaS provides fully functional software that users access directly.
- XaaS is a general term covering all cloud-based services, including Storage-as-a-Service, Database-as-a-Service, AI-as-a-Service, and more.

Conclusion

The four cloud service models address different user needs. IaaS offers infrastructure, PaaS provides a development platform, SaaS delivers ready-to-use software, and XaaS extends cloud computing to almost any IT service, making cloud solutions flexible and scalable for businesses and individuals.